

Question	Answer	Mark	Mark scheme	
23	$\frac{10\sqrt{3} + 27}{39}$	M1	for method to rationalise one of the fractions, eg $\frac{3\sqrt{3}}{4-\sqrt{3}} \times \frac{4+\sqrt{3}}{4+\sqrt{3}} (= \frac{12\sqrt{3}+9}{16-3})$ or $\frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} (= \frac{2\sqrt{3}}{3})$ oe	for method to write fractions with a common denominator, eg $\frac{3\sqrt{3} \times \sqrt{3}}{\sqrt{3}(4-\sqrt{3})} - \frac{2 \times (4-\sqrt{3})}{\sqrt{3}(4-\sqrt{3})}$ oe
		M1	for method to rationalise both of the fractions eg $\frac{3\sqrt{3}}{4-\sqrt{3}} \times \frac{4+\sqrt{3}}{4+\sqrt{3}} (= \frac{12\sqrt{3}+9}{16-3})$ and $\frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} (= \frac{2\sqrt{3}}{3})$ oe	for writing as a correct single fraction without brackets, eg $\frac{9-8+2\sqrt{3}}{4\sqrt{3}-3}$ oe
		M1	for method to write correct fractions with a common denominator eg $\frac{3(12\sqrt{3}+9)}{13 \times 3} - \frac{13 \times 2\sqrt{3}}{13 \times 3}$ or $\frac{36\sqrt{3}+27}{39} - \frac{26\sqrt{3}}{39}$ oe	for method to rationalise a fraction of the form $\frac{a+b\sqrt{3}}{c\sqrt{3}-d}$ where a, b, c and d are integers, eg $\frac{1+2\sqrt{3}}{4\sqrt{3}-3} \times \frac{4\sqrt{3}+3}{4\sqrt{3}+3}$ or $\frac{4\sqrt{3}+3+24+6\sqrt{3}}{48+12\sqrt{3}-12\sqrt{3}-9}$ oe
		A1	for $\frac{10\sqrt{3}+27}{39}$ oe	